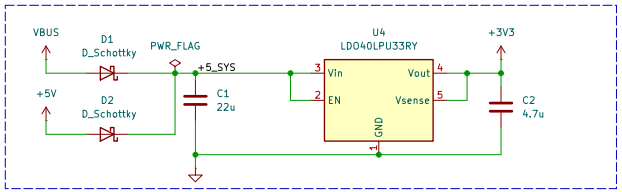
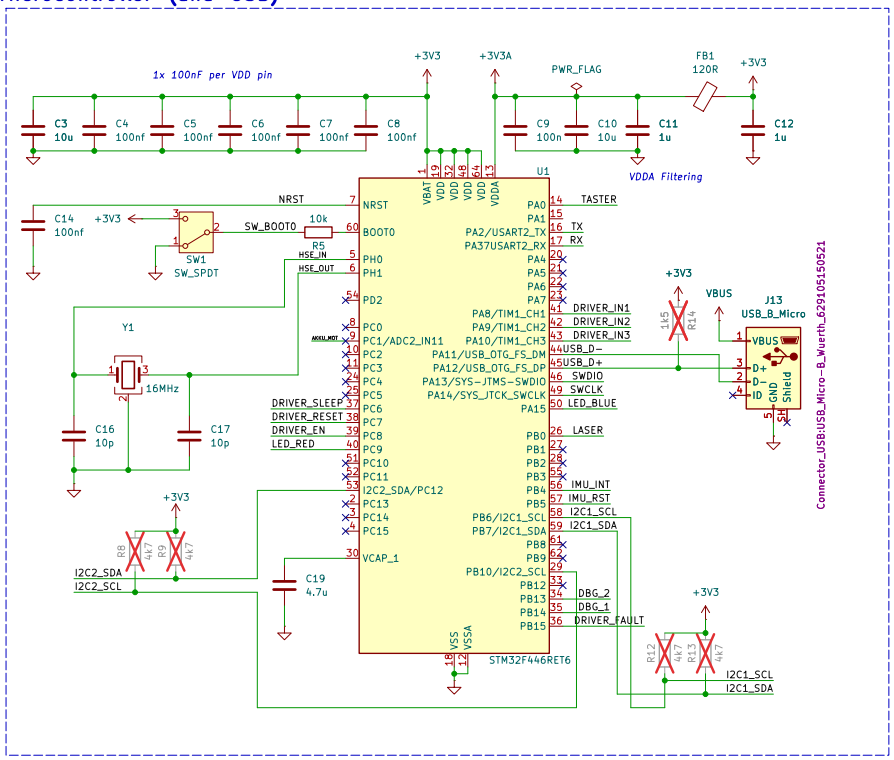
Prototype 5 STM32 PCB

Power Supply

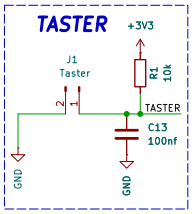


Microcontroller (and USB)

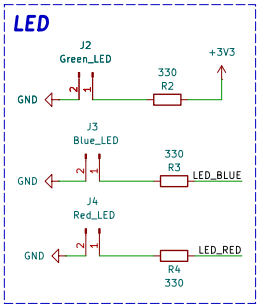


Pin-out using STM32CubeMX

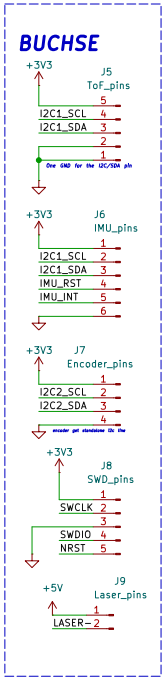
TASTER



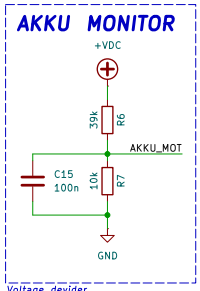
LED



BUCHSE

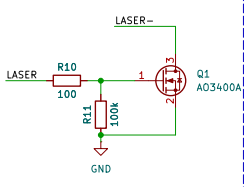


AKKU MONITOR



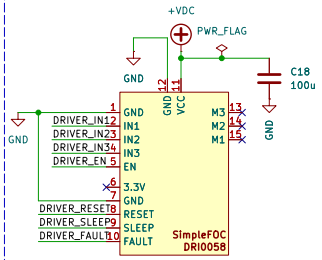
Voltage devider
V_adc = V_akku x (10k/10k + 39k)

LASER



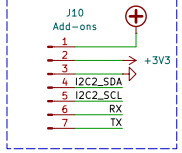
using N-Channel MOSFET as the laser switch

TREIBER



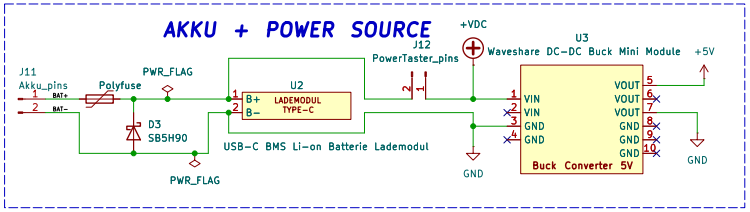
M1,M2,M3 connects directly to the Motor

ADD-ONS

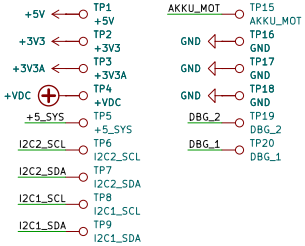


The addon part can be used for hacking the device and adding custom extensions, or while debugging. It exposes 3.3V power, the I2C bus and the RX/TX lines for serial output and communications. Test points provide solderable connections to more GPIO pins as well.

AKKU + POWER SOURCE



TEST POINTS



MOUNTING HOLES



updated MCU symbol for pins belegung
footprint finished, DNP for I2C resistors
adding Test points and mounting holes
adding Laser switch

FH Aachen

Sheet: /
File: PCB_STM32_custom_board.kicad_sch

Title: Prototype 5 STM32 PCB

Size: A3	Date: 2026-10-25	Rev: 8
KiCad E.D.A. 10.0.3		Id: 1/1